Avni Badiwale

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Education

University of Wisconsin–Madison 3.50/4.00
B.S. Computer Science & Mathematics *Sep 2024–May 2028*

Work Experience

Garmin Olathe, Kansas
Software Engineering Intern (Aviation Department) *May 2026–August 2026*

- Built a visualization tool for electrical protection curves after an ECB blew up during hardware testing, enabling engineers to catch configuration errors before deployment.
- Drove adoption among ∼50 systems engineers by embedding the tool into the team's WPF/C# (MVVM) ground-support application for configuring safety-critical avionics power distribution units.
- Automated C# style-convention enforcement across the codebase with a Roslyn-based reordering script, eliminating ∼5 hours of manual cleanup per release cycle.

Epic Systems Verona, Wisconsin
Student Assistant *May 2025–Jan 2026*

- Proposed and delivered a campus-wide indoor digital twin webmap of 37 buildings for 15,000 employees, replacing a legacy PDF-based system that required manual updates for any changes.
- Built the frontend in React with the ArcGIS SDK for JavaScript, intelligently searching and rendering across tens of thousands of indexed polygons.
- Developed RESTful APIs in C# to bridge two independent backends, unifying live spatial floor plan data with enterprise SQL employee records.
- Leveraged this unified data pipeline to enable real-time room availability, employee directory lookups, and populate interactive popups for every room within the webmap.
- Unblocked homepage integration by migrating a legacy Webpack 4 build to Vite/esbuild, and cut frontend build times by ∼70%.

Google Summer of Code at OpenMS (Project Summary) Remote
Software Engineering Intern *May 2025–Sep 2025*

- Accepted into Google Summer of Code (∼5% acceptance rate) with an 18-page technical proposal to implement columnar data formats in OpenMS's C++ scientific computing stack.
- Integrated Apache Arrow and Parquet I/O within the CMake build system, significantly reducing data parsing overhead for downstream machine learning pipelines.
- Eliminated the Windows onboarding barrier for OpenMS contributors by pioneering the project's first Conda-based build environment, reducing a multi-hour manual CMake setup to a single reproducible command.
- Unblocked the entire Arrow integration on Ubuntu, macOS, and Windows by resolving incompatible nlohmann/json versions

Personal Projects

MergeNB (VS Code Extension for Jupyter Notebook Conflict Resolution) GitHub

- First-in-market resolution engine that renders Jupyter Notebook git merge conflicts as interactive, cell-level rich diffs.
- Designed a cell-matching algorithm using Hungarian assignment and a weighted cost matrix to accurately align cells.
- Built a responsive UI using React and CodeMirror 6, powered by a custom local HTTP and WebSocket server for real-time state synchronization
- Engineered a 20-test CI/CD suite with GitHub Actions, utilizing Playwright browser automation to end-to-end test live Git merge-conflict resolution and UI rendering.

Technical Skills

Languages: C#, C++, Python, TypeScript/JavaScript, Java, SQL, Bash
Frontend & Web: React, WPF, Preact, HTML/CSS
Backend & Data: .NET, RESTful APIs, Apache Arrow, Parquet, PyTorch
Infrastructure: Git, GitHub Actions (CI/CD), Docker, Linux, CMake, Conda, Playwright